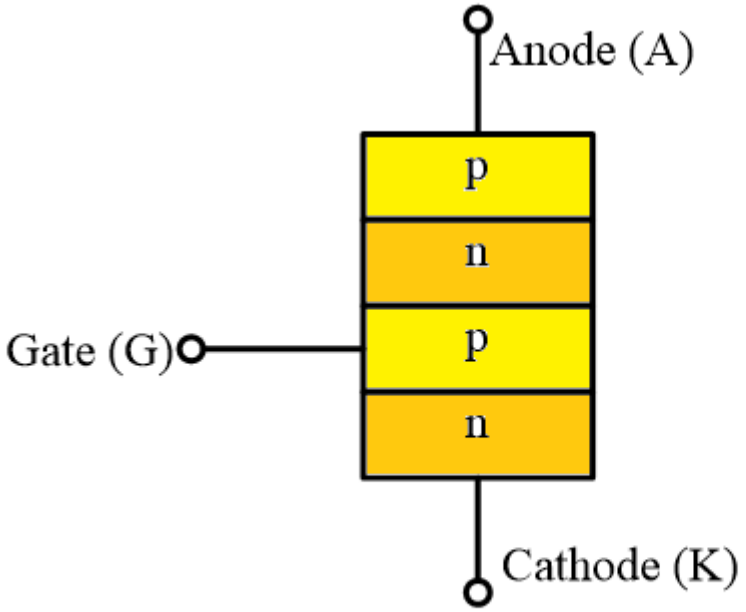
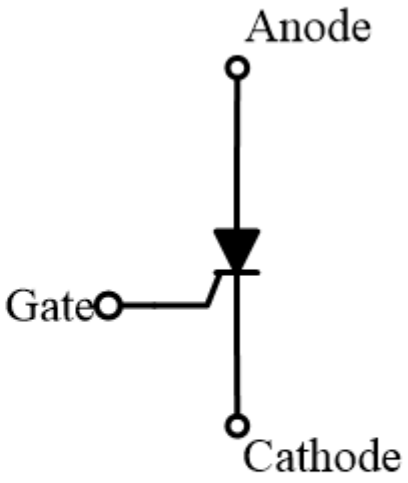
14.1 HỌ THYRISTOR

14.1.2 SCR – Silicon Controlled Rectifier

➢ ***Cấu tạo kí hiệu***



Cấu trúc
(a)



Kí hiệu
(b)

Thyristor SCR KP10A1600V Th...
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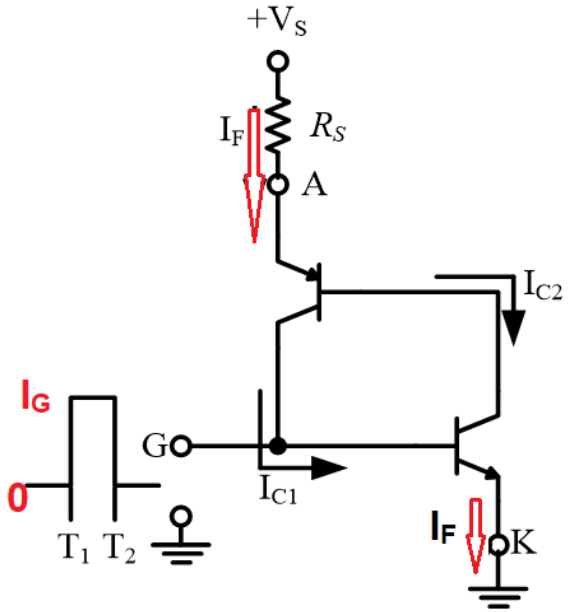
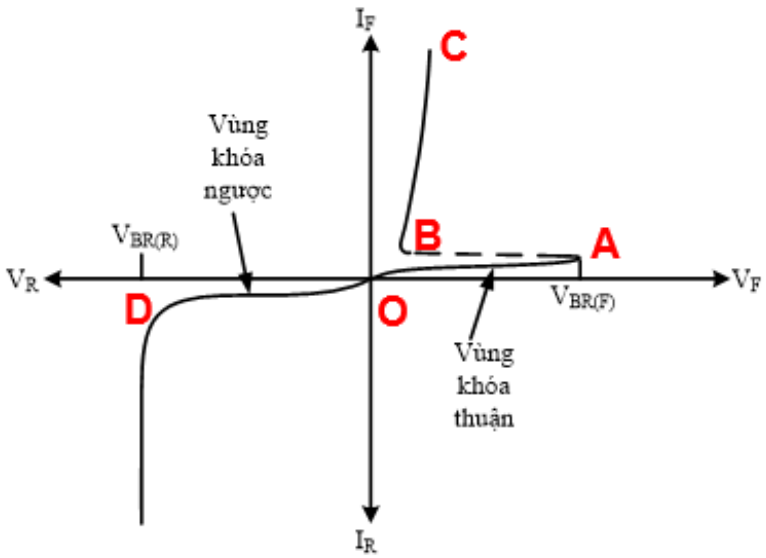
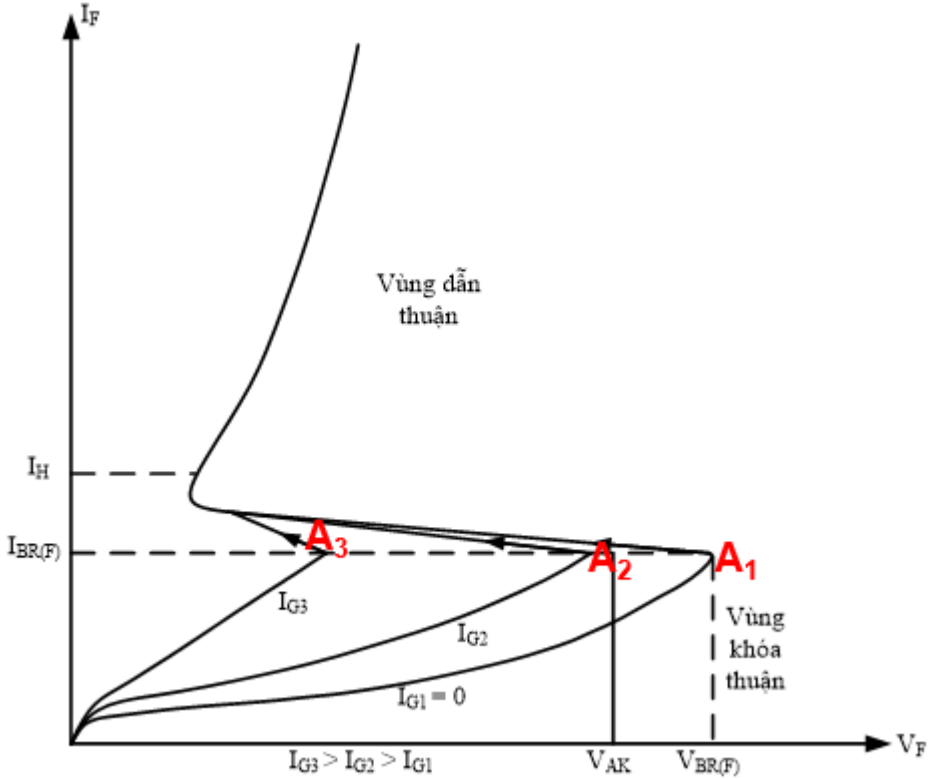
14.1 HỌ THYRISTOR

14.1.2 SCR – Silicon Controlled Rectifier

➢ ***Nguyên lý hoạt động***







- Điều kiện **dẫn**: phân cực thuận $\mathbf{V_{AK} > V_{BR(F)}}$ và kích dòng $\mathbf{I_G > 0}$.
- Điều kiện **ngắt**: giảm dòng $\mathbf{I_{AK} < I_H}$.

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14.1 HỌ THYRISTOR

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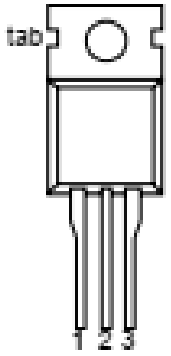

➢ ***Datasheet***

LIMITING VALUES

Limiting values in accordance with the Absolute Maximum System (IEC 134).

SYMBOL	PARAMETER	CONDITIONS	MIN.	MAX.			UNIT
				-500R	-650R	-800R	
V_{DRM}, V_{RRM}	Repetitive peak off-state voltages		-	500[1]	650[1]	800	V
$I_{T(AV)}$	Average on-state current	half sine wave; $T_{mb} \leq 109$ °C	-	7.5			A
$I_{T(RMS)}$	RMS on-state current	all conduction angles	-	12			A
I_{TSM}	Non-repetitive peak on-state current	half sine wave; T_j = 25 °C prior to surge					
		t = 10 ms	-	100			A
		t = 8.3 ms	-	110			A
I^2t	I^2t for fusing	t = 10 ms	-	50			A^2s
dI_T/dt	Repetitive rate of rise of on-state current after triggering	I_{TM} = 20 A; I_G = 50 mA; dI_G/dt = 50 mA/µs	-	50			A/µs
I_{GM}	Peak gate current		-	2			A
V_{GM}	Peak gate voltage		-	5			V
V_{RGM}	Peak reverse gate voltage		-	5			V
P_{GM}	Peak gate power		-	5			W
$P_{G(AV)}$	Average gate power	over any 20 ms period	-	0.5			W
T_{stg}	Storage temperature		-40	150			°C
T_j	Operating junction temperature		-	125			°C

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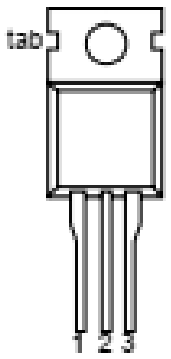

➢ ***Datasheet***

STATIC CHARACTERISTICS

T_j = 25 °C unless otherwise stated

SYMBOL	PARAMETER	CONDITIONS	MIN.	TYP.	MAX.	UNIT
I_{GT}	Gate trigger current	V_D = 12 V; I_T = 0.1 A	-	2	15	mA
I_L	Latching current	V_D = 12 V; I_{GT} = 0.1 A	-	10	40	mA
I_H	Holding current	V_D = 12 V; I_{GT} = 0.1 A	-	7	20	mA
V_T	On-state voltage	I_T = 23 A	-	1.4	1.75	V
V_{GT}	Gate trigger voltage	V_D = 12 V; I_T = 0.1 A	-	0.6	1.5	V
		V_D = $V_{DRM(max)}$; I_T = 0.1 A; T_j = 125 °C	0.25	0.4	-	V
I_D, I_R	Off-state leakage current	V_D = $V_{DRM(max)}$; V_R = $V_{RRM(max)}$; T_j = 125 °C	-	0.1	0.5	mA

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14.1 HỌ THYRISTOR

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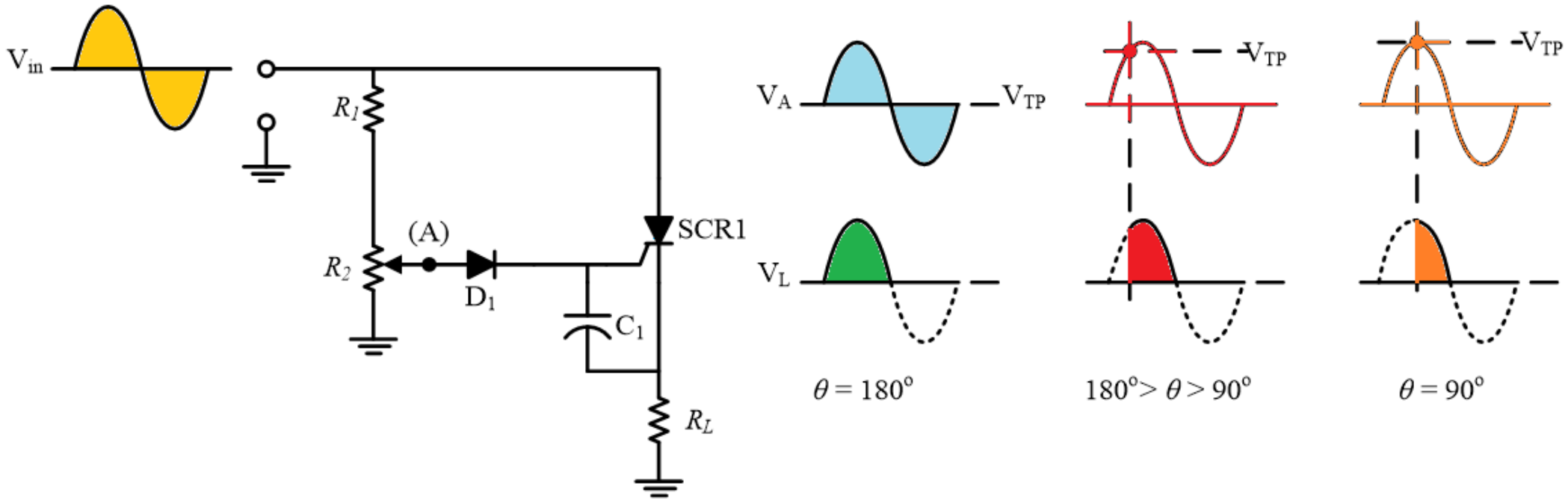
➢ ***Mạch ứng dụng***



- Vặn biến trở → thay đổi thời điểm kích → thay đổi điện áp trung bình trên tải.

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